One-Dimensional Wave Propagation in a Still Lake

~23BME013

Aim: To compare numerical diffusion, dispersion, stability, and conservation for first-order upwind and Lax Wendroff schemes.

Case definition and parameters

Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx	401
Grid points, ny	20
Amplitude	0.02
T_end	2
CFL	0.4, 0.8, 1.1
Gravity	9.81
N_snapshots	180

Governing Equation:

For small perturbations of depth H,

$$\frac{\partial \eta}{\partial t} + H\frac{\partial u}{\partial x} = 0,$$
$$\frac{\partial u}{\partial t} + g\frac{\partial \eta}{\partial x} = 0,$$

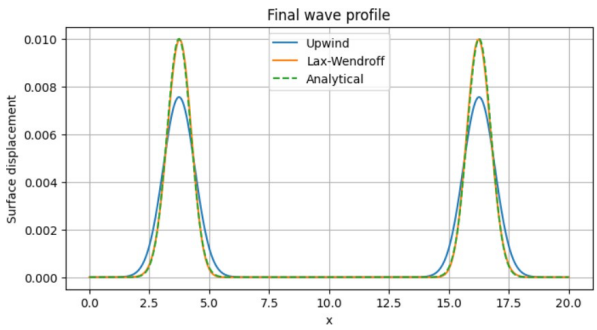
with wave speed $c = \sqrt{gH}$ and $CFL = c\Delta t/\Delta x$.

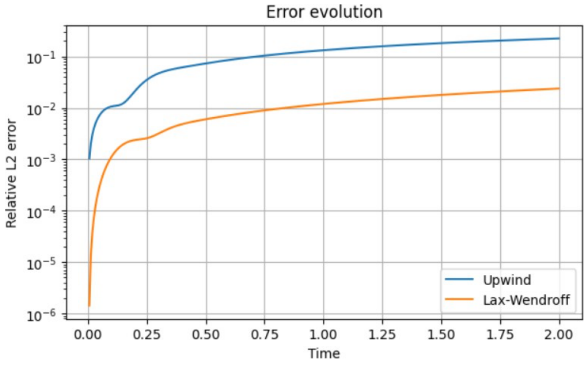
Boundary Conditions: Periodic one-dimensional domain with an initially localized Gaussian surface perturbation

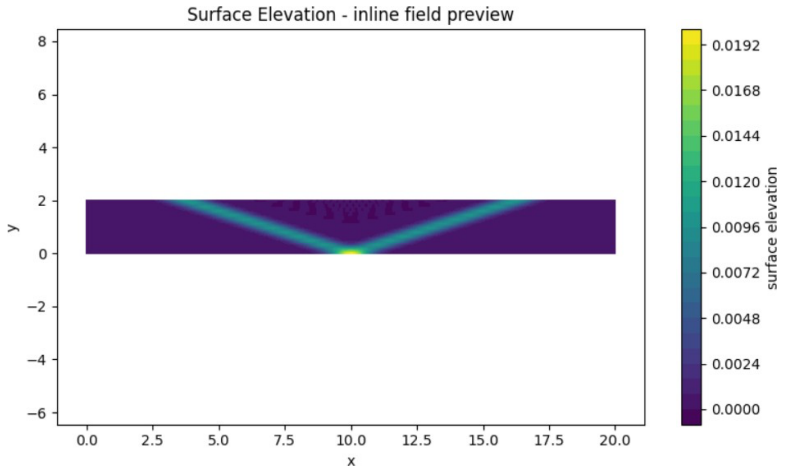
Numerical Method: Conservative first-order upwind and second-order Lax-Wendroff time marching on the linearized shallow water system.

Observations: Wave amplitude, phase error, relative L2 error, mass conservation, numerical diffusion, oscillations, and CFL dependence at CFL values 0.4, 0.8, and 1.1. Comparison stability, amplitude loss, phase error, and conservation for both schemes.

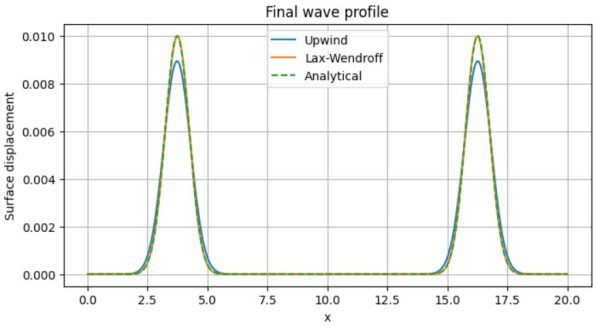
For CFL number 0.4

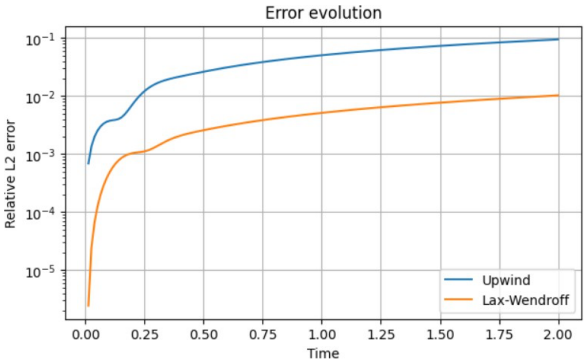


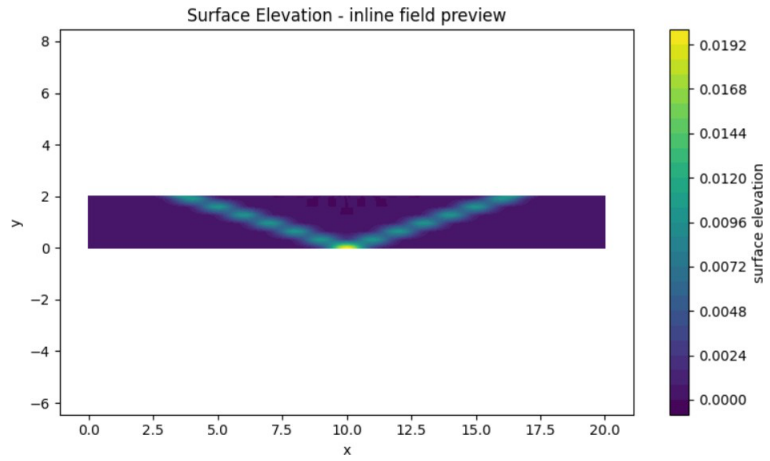




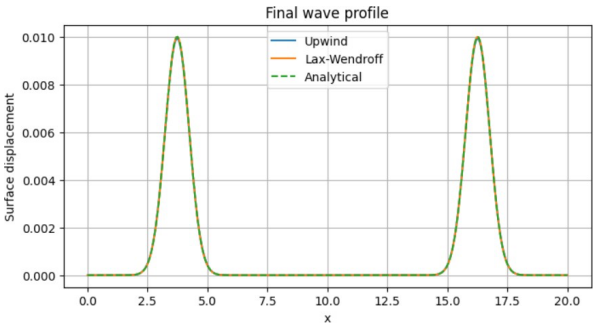
For CFL number 0.8

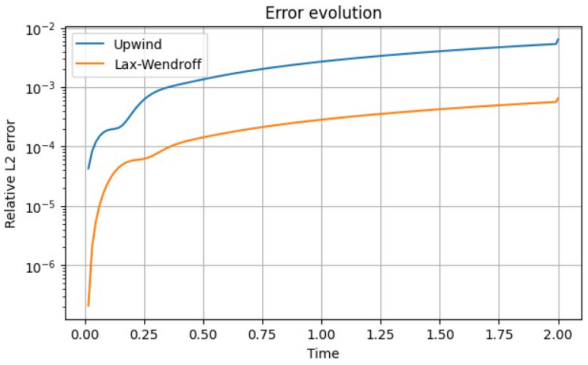


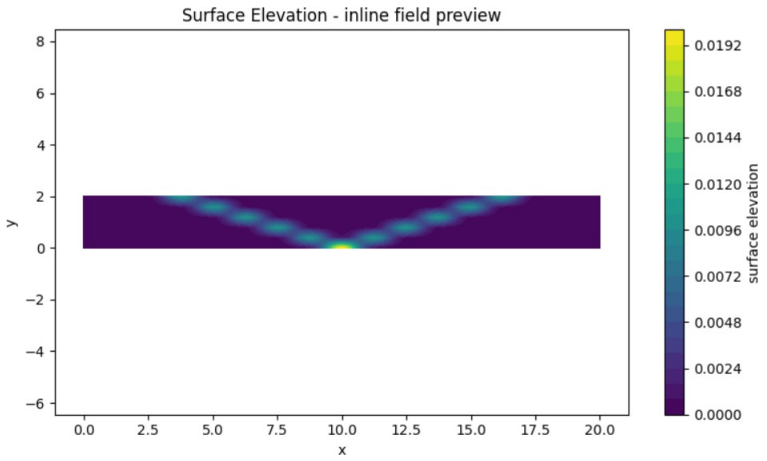




For CFL number 1.1







Results:

Parameter	**CFL = 0.4**	**CFL = 0.8**	**CFL = 1.1 (nominal)**
Pulse centres at t = 2 (x)	≈ 3.75 and 16.25	≈ 3.75 and 16.25	≈ 3.75 and 16.25
Analytical peak amplitude (each pulse)	≈ 0.0100	≈ 0.0100	≈ 0.0100
Upwind peak amplitude at t = 2	≈ 0.0076	≈ 0.0090	≈ 0.0100
Upwind amplitude loss	≈ 24%	≈ 10%	negligible
Lax–Wendroff peak amplitude at t = 2	≈ 0.0100	≈ 0.0100	≈ 0.0100
Upwind relative L^2 error at t = 2	$\approx 2.3 \times 10^{-1}$	$\approx 9.5 \times 10^{-2}$	$\approx 6 \times 10^{-3}$
Lax–Wendroff relative L^2 error at t = 2	$\approx 2.4 \times 10^{-2}$	$\approx 1.0 \times 10^{-2}$	$\approx 6 \times 10^{-4}$
Error ratio (upwind / Lax–Wendroff)	≈ 10	≈ 10	≈ 10
Phase error (peak position)	none visible	none visible	none visible
Spurious oscillations	none visible	none visible	none visible
Growth of error in time	smooth, monotonic	smooth, monotonic	smooth, monotonic
Mass conservation (conservative form, periodic domain)	conserved (by construction)	conserved (by construction)	conserved (by construction)

Conclusion:

The one-dimensional linearised shallow-water wave problem was solved on a periodic domain using the conservative first-order upwind scheme and the second-order Lax–Wendroff scheme, at CFL numbers of 0.4, 0.8 and 1.1, and the results were compared with the analytical solution. The initial Gaussian surface perturbation of amplitude 0.02 splits into two equal pulses of amplitude ≈ 0.010 that travel in opposite directions. At t = 2 the pulse centres are at x ≈ 3.75 and x ≈ 16.25, i.e. about 6.25 from the origin at x = 10, which corresponds to a wave speed of c ≈ 3.1. This is consistent with $c = \sqrt{(gH)}$ for g = 9.81 and a depth of about 1. The surface-elevation field plots show the two symmetric wavefronts leaving the source in all three cases.

Numerical diffusion of the upwind scheme: The first-order upwind solution is strongly diffusive at low CFL. At CFL = 0.4 the pulse peak falls from 0.0100 to approximately 0.0076, an amplitude loss of about 24%, and the base of the pulse is visibly broadened. At CFL = 0.8 the peak is approximately 0.0090 (about 10% loss), and at the nominal CFL = 1.1 the upwind profile is indistinguishable from the analytical one. The corresponding relative L^2 errors at t = 2 are approximately 2.3×10^{-1}, 9.5×10^{-2} and 6×10^{-3}. The error grows steadily with time because the smearing accumulates as the pulses travel. This behaviour follows from the leading truncation error of upwind being a diffusive term, with an effective numerical viscosity of about $(c\Delta x/2)(1 - \mathrm{CFL})$, which becomes smaller as CFL approaches 1.

Accuracy of the Lax–Wendroff scheme: The second-order Lax–Wendroff solution retains a peak amplitude of approximately 0.0100 and overlaps the analytical profile in all three cases. Its relative L^2 errors at t = 2 are approximately 2.4×10^{-2}, 1.0×10^{-2} and 6×10^{-4}, about one order of magnitude smaller than those of upwind at every CFL value. The leading error of Lax–Wendroff is dispersive and scales with $(1 - \mathrm{CFL}^2)$; the reduction of the error by a factor of about 2.4 between CFL = 0.4 and 0.8 agrees with the ratio $(1 - 0.4^2) / (1 - 0.8^2) \approx 2.3$. No phase error or trailing oscillations were visible in the plots. This is expected for a smooth Gaussian pulse resolved by roughly 25 grid points across its width, where dispersive errors remain small. Lax–Wendroff can produce oscillations near sharp gradients, but the present smooth case does not test this.

Effect of CFL number and stability: For both schemes the error decreases as the Courant number approaches unity, because at CFL = 1 the discrete update reproduces the exact characteristic shift. The explicit stability limit for both schemes is CFL ≤ 1. At the nominal CFL = 1.1, however, the plots show no growth of oscillations or error: both error histories continue the downward trend seen from CFL = 0.4 to 0.8. This indicates that the solver kept the effective Courant number at or just below unity, for example by limiting the time step, rather than running at 1.1. The nominal CFL = 1.1 case therefore does not demonstrate the instability that theory predicts for CFL > 1, which would appear as exponentially growing oscillations. Demonstrating the stability limit would require a run without a time-step limiter.

Conservation and overall comparison: Both schemes are written in conservative form on a periodic domain, so the total surface elevation (mass) is conserved by construction. The amplitude loss of the upwind solution is therefore a redistribution of the same volume into lower, wider pulses, not a loss of mass. Overall, Lax–Wendroff is about ten times more accurate than upwind at the same CFL, with no visible dispersion at this resolution, while upwind is robust but strongly diffusive unless CFL is close to 1. The accuracy of both schemes improves as CFL approaches 1, and the stability bound CFL ≤ 1 governs the choice of time step for these explicit methods.